Lab-04 Report: Generated 2026-10-12 19:52

Names:

Group Name:

- **Q1.1:** In the code block below, explore both the training and validation sets of the CIFAR-10 dataset. Just like with the MNIST dataset, when using a torchvision dataset you can use the `.data` and `.targets` attributes to access the data arrays and labels. What are some ways in which this data is similar to what we previously worked with (how is it similar in nature to MNIST)? In what ways is it different? What makes it "more complex"? Be specific.
- **A1.1:** This is similar as it is images (obviously) but not they contain color and are effectively downsampled versions of images until they are 32 x 32 x 3. This means that the images are 32x32 instead of 28x28 and the last dimension (which before was pixel intensity) are now color channels! The dataset is large at 50,000 training images and 10,000 validation images.

```
print(cifar10_train)

# YOUR CODE STARTS HERE
print(cifar10_train.data)
print(cifar10_train.data.shape)
print(f"train shape: {cifar10_train.targets}")



print(f"val shape: {cifar10_val.data.shape}")
print(cifar10_val.targets)
plt.imshow(cifar10_train.data[1])
plt.title(f"Label: {cifar10_train.targets[1]}")
plt.show()
plt.imshow(cifar10_val.data[1])
plt.title(f"Label: {cifar10_val.targets[1]}")
plt.show()
# YOUR CODE ENDS HERE
```

```
Dataset CIFAR10
    Number of datapoints: 50000
    Root location: ./data
    Split: Train
[[[[ 59  62  63]
   [ 43  46  45]
   [ 50  48  43]
   ...
   [158 132 108]
   [152 125 102]
   [148 124 103]]

  [[ 16  20  20]
   [  0   0   0]
   [ 18   8   0]
   ...
   [123  88  55]
   [119  83  50]
   [122  87  57]]

  [[ 25  24  21]
   [ 16   7   0]
   [ 49  27   8]
   ...
   [118  84  50]
   [120  84  50]
   [109  73  42]]

  ...

  [[208 170  96]
   [201 153  34]
   [198 161  26]
   ...
   [160 133  70]
   [ 56  31   7]
   [ 53  34  20]]

  [[180 139  96]
   [173 123  42]
   [186 144  30]
   ...
   [184 148  94]
   [ 97  62  34]
   [ 83  53  34]]

  [[177 144 116]
   [168 129  94]
   [179 142  87]
   ...
   [216 184 140]
   [151 118  84]
   [123  92  72]]]
```

```
[[[154 177 187]
  [126 137 136]
  [105 104  95]
  ...
  [ 91  95  71]
  [ 87  90  71]
  [ 79  81  70]]

 [[140 160 169]
  [145 153 154]
  [125 125 118]
  ...
  [ 96  99  78]
  [ 77  80  62]
  [ 71  73  61]]

 [[140 155 164]
  [139 146 149]
  [115 115 112]
  ...
  [ 79  82  64]
  [ 68  70  55]
  [ 67  69  55]]

 ...

 [[175 167 166]
  [156 154 160]
  [154 160 170]
  ...
  [ 42  34  36]
  [ 61  53  57]
  [ 93  83  91]]

 [[165 154 128]
  [156 152 130]
  [159 161 142]
  ...
  [103  93  96]
  [123 114 120]
  [131 121 131]]

 [[163 148 120]
  [158 148 122]
  [163 156 133]
  ...
  [143 133 139]
  [143 134 142]
  [143 133 144]]]
```

```
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  [254 254 254]]

 ...

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  [111 118 111]
  [105 112 106]
  ...
  [ 72  81  80]
  [ 72  80  79]
  [ 72  80  79]]

 [[111 118 110]
  [104 111 104]
  [ 99 106  98]
  ...
  [ 68  75  73]
  [ 70  76  75]
  [ 78  84  82]]

 [[106 113 105]
  [ 99 106  98]
  [ 95 102  94]
  ...
  [ 78  85  83]
  [ 79  85  83]
  [ 80  86  84]]]


...
```

```
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  [ 40 176 239]
  [ 42 176 241]
  ...
  [ 99 177 219]
  [ 79 147 197]
  [ 89 148 189]]

 [[ 57 182 234]
  [ 44 184 250]
  [ 50 183 240]
  ...
  [156 182 200]
  [141 177 206]
  [116 149 175]]

 [[ 98 197 237]
  [ 64 189 252]
  [ 69 192 245]
  ...
  [188 195 206]
  [119 135 147]
  [ 61  79  90]]

 ...

 [[ 73  79  77]
  [ 53  63  68]
  [ 54  68  80]
  ...
  [ 17  40  64]
  [ 21  36  51]
  [ 33  48  49]]

 [[ 61  68  75]
  [ 55  70  86]
  [ 57  79 103]
  ...
  [ 24  48  72]
  [ 17  35  53]
  [  7  23  32]]

 [[ 44  56  73]
  [ 46  66  88]
  [ 49  77 105]
  ...
  [ 27  52  77]
  [ 21  43  66]
  [ 12  31  50]]]


[[[189 211 240]
```

```
  [186 208 236]
  [185 207 235]
  ...
  [175 195 224]
  [172 194 222]
  [169 194 220]]

 [[194 210 239]
  [191 207 236]
  [190 206 235]
  ...
  [173 192 220]
  [171 191 218]
  [167 190 216]]

 [[208 219 244]
  [205 216 240]
  [204 215 239]
  ...
  [175 191 217]
  [172 190 216]
  [169 191 215]]

 ...

 [[207 199 181]
  [203 195 175]
  [203 196 173]
  ...
  [135 132 127]
  [162 158 150]
  [168 163 151]]

 [[198 190 170]
  [189 181 159]
  [180 172 147]
  ...
  [178 171 160]
  [175 169 156]
  [175 169 154]]

 [[198 189 173]
  [189 181 162]
  [178 170 149]
  ...
  [195 184 169]
  [196 189 171]
  [195 190 171]]]


[[[229 229 239]
  [236 237 247]
  [234 236 247]
```

```
   ...
   [217 219 233]
   [221 223 234]
   [222 223 233]]

  [[222 221 229]
   [239 239 249]
   [233 234 246]
   ...
   [223 223 236]
   [227 228 238]
   [210 211 220]]

  [[213 206 211]
   [234 232 239]
   [231 233 244]
   ...
   [220 220 232]
   [220 219 232]
   [202 203 215]]

  ...

  [[150 143 135]
   [140 135 127]
   [132 127 120]
   ...
   [224 222 218]
   [230 228 225]
   [241 241 238]]

  [[137 132 126]
   [130 127 120]
   [125 121 115]
   ...
   [181 180 178]
   [202 201 198]
   [212 211 207]]

  [[122 119 114]
   [118 116 110]
   [120 116 111]
   ...
   [179 177 173]
   [164 164 162]
   [163 163 161]]]]
(50000, 32, 32, 3)
train shape: [6, 9, 9, 4, 1, 1, 2, 7, 8, 3, 4, 7, 7, 2, 9, 9, 9, 3, 2, 6, 4,
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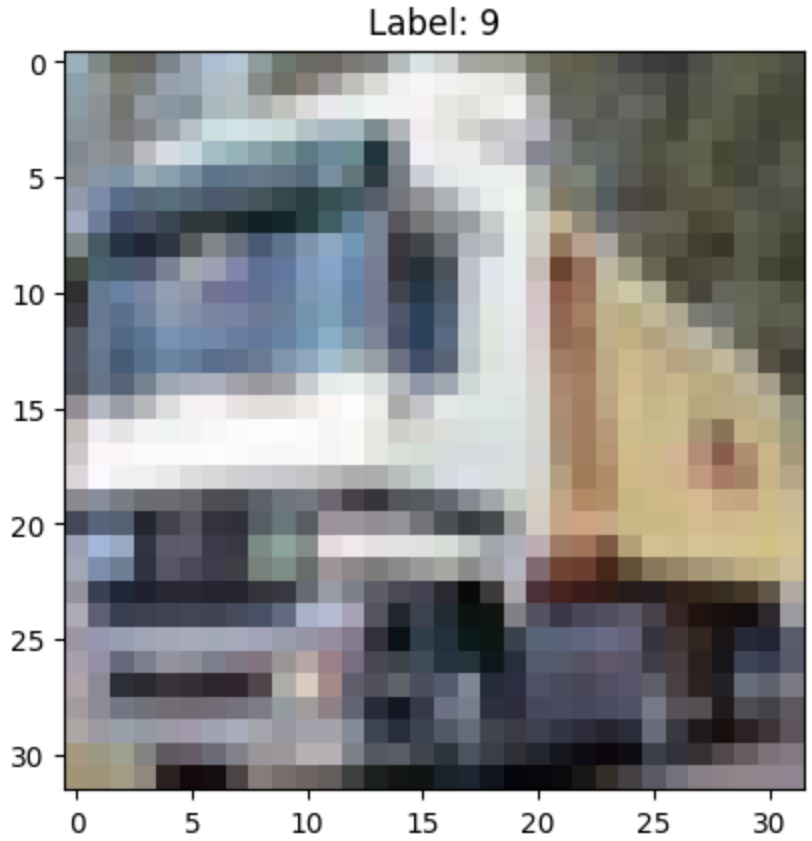
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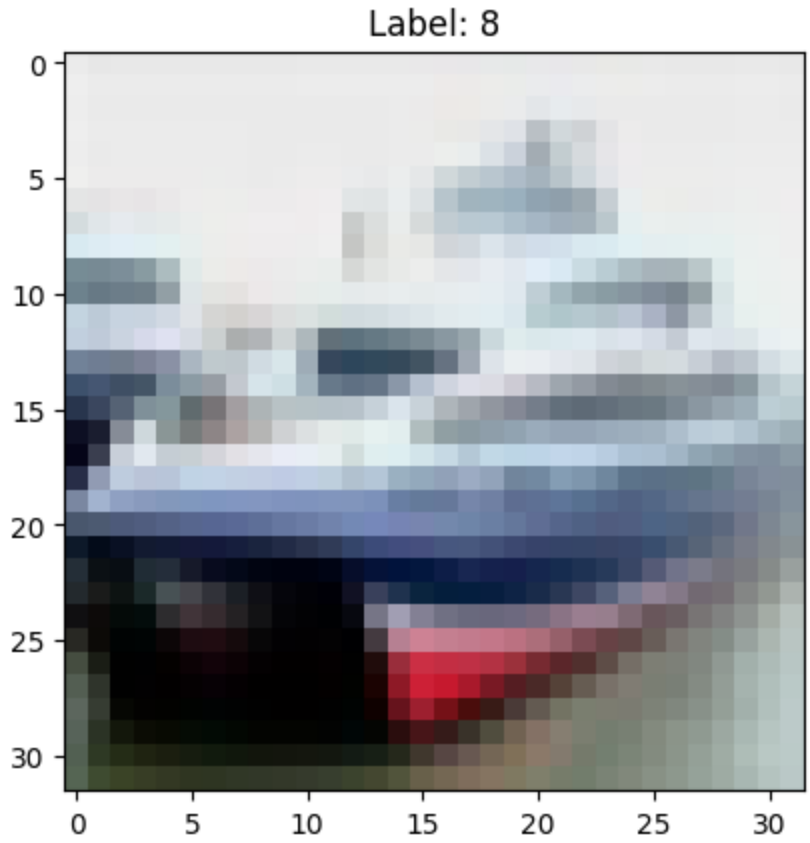
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7, 5, 8, 0, 8, 2, 7, 0, 3, 5, 3, 8, 3, 5, 1, 7]
```






```
targets = np.array(cifar10_train.targets)
images = np.array(cifar10_train.data)

fig, axes = plt.subplots(10, 10, figsize=(22, 22))
fig.suptitle("CIFAR-10 sample images (10 per class)", fontsize=32)

for row, class_name in enumerate(cifar10_labels):

    print(row, class_name)

    # Pick 10 random indices for this class
    class_idx = np.where(targets == row)[0]
    sample_idx = np.random.choice(class_idx, size=10, replace=False)

    print(sample_idx)

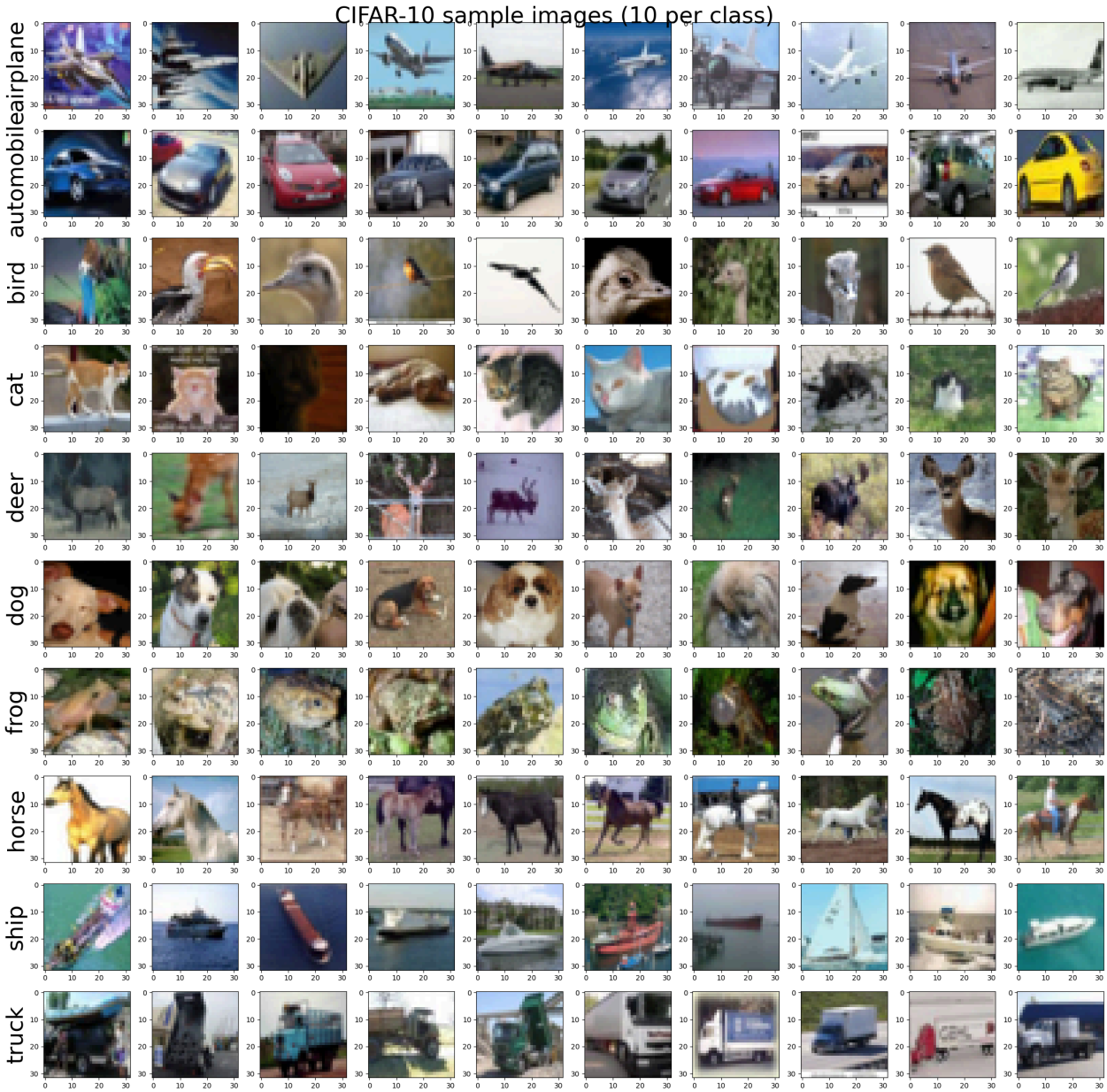
# YOUR CODE STARTS HERE
    for col in range(10):
        ax = axes[row, col]
        ax.imshow(images[sample_idx[col]])


# YOUR CODE ENDS HERE

    axes[row, 0].set_ylabel(class_name, fontsize=32)
```

```
plt.tight_layout()
plt.show()
```

```
0 airplane
[ 6373 11238 42493 15887 16214 19828 27609 32107  4868 21758]
1 automobile
[13697 35688 47704  5635 37455 44625 12676 16800 46323 40759]
2 bird
[31294  3796  6011 20318 11187 32430 13985  3339 42045 47940]
3 cat
[17835 45948 44226 37425 30159 41330 44313 19388 41334 12552]
4 deer
[41031 49252 19936 12301  9931 26649 21868 11435 33590 26118]
5 dog
[37229 48775 35606   337 32425 10423 26280  4319 41190 35044]
6 frog
[28171 24888  8033 26387 34932 32869 10195  8170 44486 47764]
7 horse
[34392 19391 29592 11527  3104 25791 22201 13039  4059 47129]
8 ship
[10775 48668 33533  5217 23838 27839 39350 18307   353 49764]
9 truck
[21507  7461  7628  5332 40810 29352 46629 13867 12568 44996]
```



- **Q1.2):** Based off of the mean image values per class, are there some classes a model may struggle to differentiate between? Give some examples of classes with similar patterns. Begin hypothesizing; how could our model still have success when some classes have such similar patterns?

- **A1.2):** It might struggle against deer and horse, as they share a similar background and general shape. While from the mean image ship and airplane look the same (blue background) the model would probably still succeed because the individual ship and airplane are likely much different.

```
class CIFAR10Dataset(d2l.DataModule):
    def __init__(self, batch_size=32, num_workers=4, resize=(32, 32)):
```

```
        super().__init__()
        self.save_hyperparameters()

        trans = None
        # Use transforms.Compose to resize, convert to tensors in the range
[0-1], and normalize on a channel-by-channel basis

# YOUR CODE STARTS HERE
        trans = transforms.Compose([
        transforms.Resize(resize),
        transforms.ToTensor(),
        transforms.Normalize(
            mean=(0.4914, 0.4822, 0.4465),
            std=(0.2470, 0.2435, 0.2616),
            ),
        ])
# YOUR CODE ENDS HERE

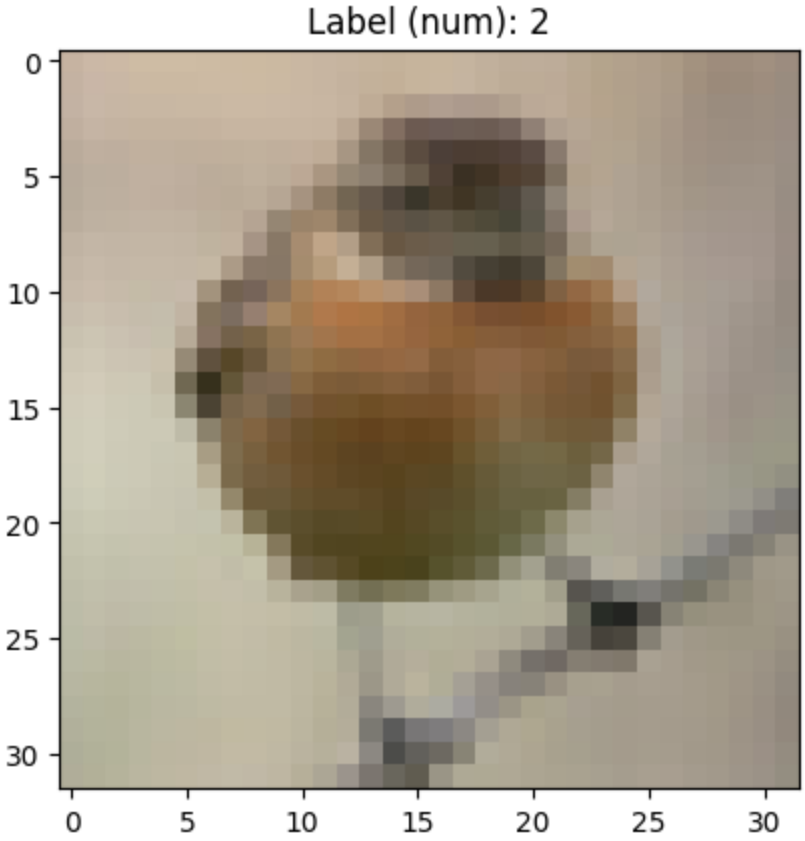
        self.train = torchvision.datasets.CIFAR10(root='./data', train=True,
download=True, transform=trans)
        self.val = torchvision.datasets.CIFAR10(root='./data', train=False,
download=True, transform=trans)


    def get_dataloader(self, train):
        data = self.train if train else self.val
        return torch.utils.data.DataLoader(data, self.batch_size,
shuffle=train,
                                           num_workers=self.num_workers)
```

```
# Instantiate the CIFAR10Dataset and inspect the first batch of the training
data

# YOUR CODE STARTS HERE
CIFAR_Data = CIFAR10Dataset()
X, y = next(iter(CIFAR_Data.get_dataloader(train=True)))
mean = torch.tensor((0.4914, 0.4822, 0.4465)).view(3, 1, 1)
std = torch.tensor((0.2470, 0.2435, 0.2616)).view(3, 1, 1)

image = X[0] * std + mean # IF WE DONT UNNORMALIZE WE SEE A SATANIC IMAGE!!!!
#print(f"x: {X}")
plt.imshow(image.permute(1,2,0))
plt.title(f"Label (num): {y[0]}")
plt.show()
# YOUR CODE ENDS HERE
```



- **Q1.3):** Why did this step take significantly less time to execute than our previous cell that loaded the dataset?
- **A1.3):** Because we only took a batch instead of the full 60k image dataset.

- **Q1.4):** How has the resulting tensor shape and values for each individual image changed from when we explored the images in the dataset directly earlier? Which function reshaped the images? Be specific.
- **A1.4):** It got reshaped into 3x32x32 and got normalized (between 0-1). The specific function that reshaped the image was the transform.compose, which also normalized it. After unnormalizing and reshaping (permute) we got a cute kitty instead of a strange satanic-esque image.

- **Q1.5):** Did the linear model perform better or worse than you expected? Compare its performance to what you previously achieved on MNIST.
- **A1.5):** It actually performed better than I expected, although it still sucks. We got about 28% val accuracy, which is much worse than the +95% we got on

MNIST. I would like to point out that while it is still really bad we are getting better than random guessing even with the linear model (random guessing would be ~10%).

- **Q2.1:** Explain, in your own words, why this new two-layer linear model would likely not be any more effective than our original one-layer linear model. Be specific, and refer back to the D2L book if needed.
- **A2.1:**

- **Q2.2:** Consider the positive region of the graph. What is the derivative of this region? Is it a constant value, or does it vary? Why is this derivative meaningful for our neural network?
- **A2.2:** The derivative of the positive region is just a constant 1 (power rule of X). This is good because it is easy to calculate!

```
def relu(X):
    # Create zeros array
    zeros = torch.zeros_like(X)
    #perform relu
    return torch.max(zeros, X)
```

```
class MLPClassifier_v1(d2l.Classifier):
    def __init__(self, num_inputs, num_outputs, num_hidden, lr, sigma=0.01):
        super().__init__()
        self.save_hyperparameters()

# YOUR CODE STARTS HERE
        self.W1 = torch.nn.Parameter(torch.normal(0, sigma, size=
(num_inputs,num_hidden)))
        self.W2 = torch.nn.Parameter(torch.normal(0, sigma, size=
(num_hidden,num_outputs)))
        self.b1 =
torch.nn.Parameter(torch.zeros_like(torch.Tensor(num_hidden)))
        self.b2 =
torch.nn.Parameter(torch.zeros_like(torch.Tensor(num_outputs)))
# YOUR CODE ENDS HERE
```

- **Q2.3:** Implement the `forward()` pass in the code block below, and briefly explain your code for this part.
- **A2.3:** We first reshape to flatten the HWC by batch. Then we do a matrix mult with @ and add the bias, then take the Relu to get the input for the hidden layer.

Then the final output is the matrix mult of the hidden layer plus bias.

```
@d2l.add_to_class(MLPClassifier_v1)
def forward(self, X):
        X = X.reshape(X.shape[0], -1)
        hidden = relu(X @ self.W1 + self.b1)
        return hidden @ self.W2 + self.b2
```

- **Q2.4:** For both `MLPClassifier_v1` and `MLPClassifier_v2`, how much did our accuracies improve from the linear model? Are the loss values meaningfully lower?
- **A2.4:** Our val accuracy nearly doubled from the linear model! THe loss values also decresed significantly, although the training loss continued to fall while the validation loss stalled out.

- **Q3.1:** Considering the final equation for the convolutional layer, how many nested `for` loops would you need to calculate this using raw Python loops? What would each one handle looping through? Be specific.
- **A3.1:** It appears we would need 6 nested for loops, as we have three Sums to output to the three output dimensions H_i,j,d (rows, cols, channels) and for each output we would have 3 input dims to loop over for each filter.

- **Q3.2:** Suppose you have a standard 32×32 pixel CIFAR-10 image. You slide a 3×3 kernel across it moving one pixel at a time (this is called the *stride*). What will the height and width of the output feature map be?
- **A3.2:** The output will be 30x30 as without any padding with a stride of one the 3x3 kernel combines the outer border of 1 pixel when generating the output. If you had padding of 1 then you would still have a 32x32 output.

```
def corr2d(X, K):

    #X is 2D input matrix of shape (height, width)
    #K is 2D kernel matrix of shape (height, width)
    h, w = X.shape
    kh, kw = K.shape

    # Spatial output dimension math (Assuming stride=1, padding=0)
    out_h = h - kh + 1
    out_w = w - kw + 1
```

```
    # Pre-allocate the output feature map with zeros
    Y = torch.zeros((out_h, out_w))

    # Perform element-wise multiplication of the window with K and sum the
total

# YOUR CODE STARTS HERE
    for i in range(out_h):
        for j in range(out_w):
            Y[i][j] = (X[i:i+kh, j:j+kw] * K).sum()
# YOUR CODE ENDS HERE
    return Y
```

- **Q3.3:** Run the test below to ensure that your cross correlation math is correct. Describe your debugging process, what mistakes (if any) you made while implementing `corr2D`, and how you fixed them.
- **A3.3:**

```
class MyConv2D(nn.Module):

    def __init__(self, kernel_size):
        super().__init__()

        # YOUR CODE STARTS HERE
        self.w = nn.Parameter(torch.randn(kernel_size))
        # YOUR CODE ENDS HERE
```

```
@d2l.add_to_class(MyConv2D)  #@save
def forward(self, x):
        # YOUR CODE STARTS HERE
        return corr2d(X=x, K=self.w)
        # YOUR CODE ENDS HERE
```

- **Q3.4:** Review Pytorch's documentation for Conv2D(). What features would you need to add to your `MyConv2D()` module to match `Conv2d()`'s functionality? Be specific.
- **A3.4:** Well, we only support one kernel and only do one basic MAC. If we wanted to implement the same functionality we would need to add a bias, input/output channels, stride parameters, padding parameters. Otherwise we support the core 2D cross-correlation operation.

```
def pool2d(X, pool_size=(2, 2), stride=2):
    h, w = X.shape
    ph, pw = pool_size
```

```
    # Calculate output dimensions based on the stride step
    out_h = (h - ph) // stride + 1
    out_w = (w - pw) // stride + 1

    # Initialize the downsampled output matrix
    Y = torch.zeros((out_h, out_w))

    # Find the maximum value inside this window and save it to Y[i,j] (hint:
use torch.max())

# YOUR CODE STARTS HERE
    for i in range(out_h):
        for j in range(out_w):
            Y[i][j] = torch.max(X[i*stride:i*stride+ph,j*stride:j*stride+pw])
# YOUR CODE ENDS HERE

    return Y
```

- **Q4.1:** Review the code block below, and describe how the size and shape of an input batch of CIFAR-10 data will be changed as it is processed layer-by-layer through the network. Be specific.
- **A4.1:** Based on the input of the first layer, we have 3 input channels, so we likely have RGB values. After the first layer, it will not change size because we have a 3x3 kernel and 1 padding but the output channels will grow from 3 to 16. Then the model will apply ReLU, Then it will be fed into maxpool which will take the max of every 2x2 square on the image with a stride of two, which will reduce the wiedth and height by 1/2 (so we will have 16x16x16). Then out image is finally passed through two linear fully connected layers with relu in the middle which will finally give us our 10 output predictions.

- **Q4.2:** Summarize the changes you made to achieve the best performance in the model (learning rate, layers added, epoch count, etc). What was your best training and testing accuracy?
- **A4.2:** First we added another convolutional + relu + flatten layer to make our model deeper. This resulted in the first fully connected layer having 2048 inputs instead as the last conv output is 8x8x32. It looked great, but around epoch 8 the validation loss shot back up as it overfut giving us a final accuracy of 68.8% which was just under 2% better (still an improvement). The next round we tried just 6 epochs but the same lr and layers. It ended too early at 65% so it was a bust. The next round we doubled the learning rate to 0.02, Which saw the training loss fall very rapidly at first, but the model appeared to overfit very quickly but actually had the highest validation accuracy of 71%

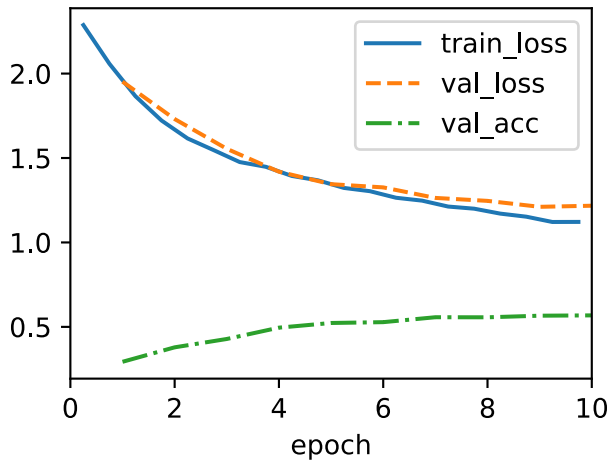
- **Q4.3:** In the code block below, create an instance of `LeNet()` with a learning rate of 0.01. Train this model on CIFAR-10 and print out the resulting training and validation results using `print_model_results()`. How does LeNet compare to your custom CNN from before? Why might this be the case?

- **A4.3:** While both our model and Le's model has 2 Conv2d layers, LeCunn's model uses Average pooling instead of maxpool. In addition to that it has 3 fully connected layers instead of our models 2. Regardless, our model had a higher validation accuracy of 71% compared to his 56%. This is maybe because our first fully connect layer has almost 2x as many inputs which might give the fully connected layers a better chance to learn more as it has more features that are less refined.

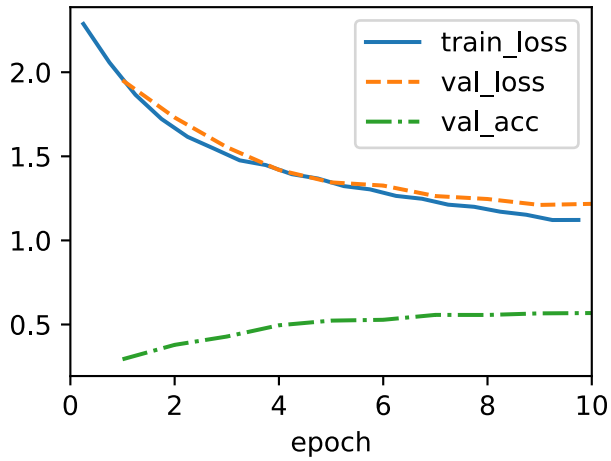
```
# Instantiate the model and trainer, and fit using the CIFAR_Data.

# YOUR CODE STARTS HERE
modelLe = LeNet(num_classes=10, lr=0.01)
model_trainerLe = d2l.Trainer(max_epochs=10)
model_trainerLe.fit(modelLe, CIFAR_Data)

print_model_results(modelLe.board)
# YOUR CODE ENDS HERE
```



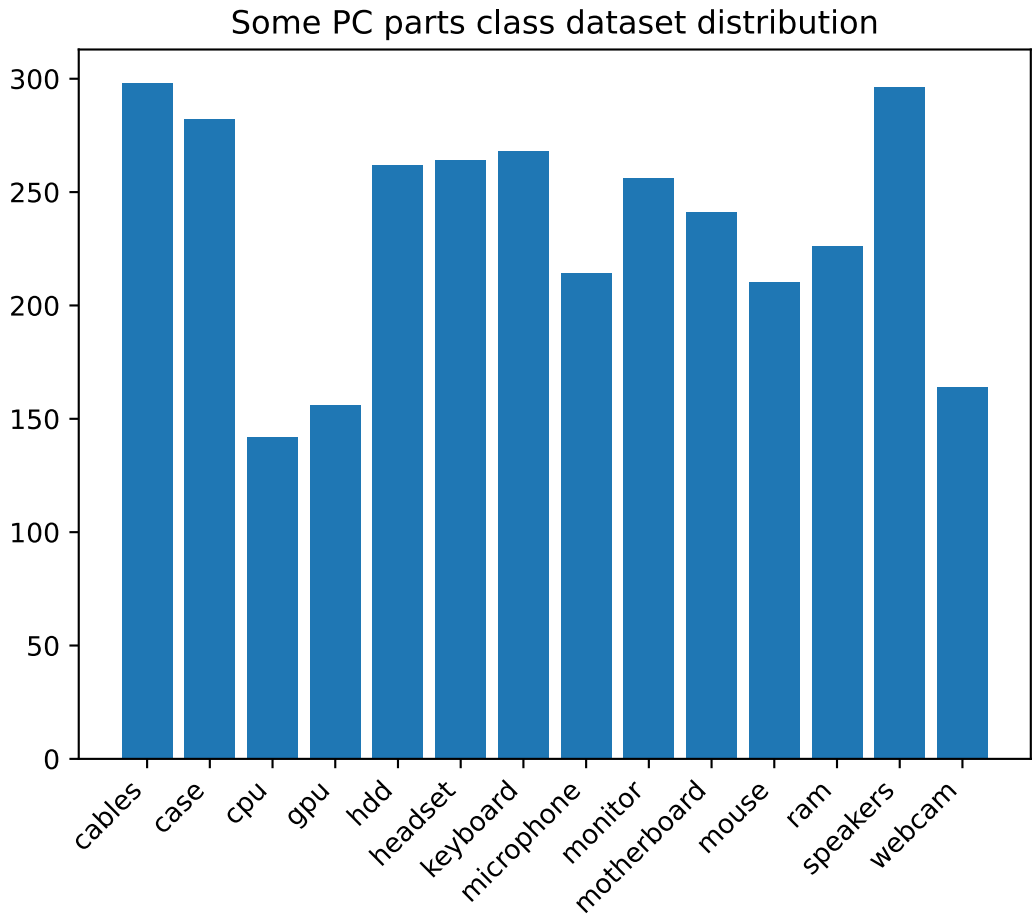

```
========================================
Final Training Loss:     1.1216
Final Validation Loss:   1.2174
----------------------------------------
Final Validation Accuracy: 56.81%
========================================
```



- **Q5.1:** How is the datatype associated with your `pcparts` variable different from the CIFAR-10 dataset? What additional steps will be required to work with `pcparts`?
- **A5.1:** There are 14 classes to predict, which is the biggest change we have. Our last dense layer will need to output 14 classes. The datatype is also a ImageFolder, which I'm not sure if this will impact how we construct and use the dataset.

```
# Count the frequencies for each label
pcparts_label_series = pd.Series(pcparts.targets)
class_counts = pcparts_label_series.value_counts().sort_index()

# YOUR CODE STARTS HERE
plt.bar(pcparts.classes, class_counts)
plt.xticks(rotation=45, ha='right')
plt.title("Some PC parts class dataset distribution")
plt.show()
# YOUR CODE ENDS HERE
```



- **Q5.2:** How would this 256×256 input size impact either of the CNN models we have previously worked with? Be specific.
- **A5.2:** 256x256 is much larger than the 32x32 we were working with. We also have 3 color channels so we are working with 256x256x3 = 196,608 input features compared to the 3,072 of the previous. This means that without reducing the dims rapidly the computational cost is much higher.

```
class PCPartsData(d2l.DataModule):

    def __init__(self, data_dir, batch_size=64, resize=(128, 128)):
        super().__init__()
        self.save_hyperparameters()

        # Use transforms.Compose() to resize and normalize images

# YOUR CODE STARTS HERE
        self.transform = transforms.Compose([transforms.Resize(resize),
transforms.ToTensor(), transforms.Normalize(mean=[0.5,0.5,0.5], std=
[0.5,0.5,0.5])])
# YOUR CODE ENDS HERE
```

```
        # Load the complete dataset from the directory structure
        self.full_dataset = torchvision.datasets.ImageFolder(
            root=data_dir,
            transform=self.transform
        )

        # 80/20 train/validation split
        train_size = int(0.8 * len(self.full_dataset))
        val_size = len(self.full_dataset) - train_size

        print(train_size, val_size)
        self.train_set, self.val_set = torch.utils.data.random_split(
            self.full_dataset,
            [train_size, val_size],
        )

        self.num_classes = len(self.full_dataset.classes)
        self.classes = self.full_dataset.classes

    def get_dataloader(self, train):
        data = self.train_set if train else self.val_set
        return torch.utils.data.DataLoader(data, self.batch_size,
shuffle=train,
                                           num_workers=self.num_workers)
```

```
# Review how the first batch looks

# YOUR CODE STARTS HERE
PCPartsDataImpl = PCPartsData(data_path, resize=(224,224))
X, y = next(iter(PCPartsDataImpl.get_dataloader(True)))
print(f"X shape:  {X.shape}")
print(f"y shape:  {y.shape}")
# YOUR CODE ENDS HERE
```

```
2623 656
X shape:  torch.Size([64, 3, 224, 224])
y shape:  torch.Size([64])
```

- **Q5.3:** Explain each line of the code you added to the `CNNClassifier_v3` `implementation below. Be specific.
- **A5.3:** First line is setting up our neural net. Line two is the first conv layer, outputing 16 channels, then we relu and maxpool. Then conv2d, with 32 output channels, relu again, maxpool, flatten into a 1D tensor, pass it into the first dense layer (lazy because I got sick of recalculating the input dims every time I made a change) then relu one more time before the last linear layer.

```
class CNNClassifier_v3(d2l.Classifier):
    def __init__(self, num_classes, lr=0.001):
```

```
        super().__init__()
        self.save_hyperparameters()

# YOUR CODE STARTS HERE
        self.net = nn.Sequential(
        nn.Conv2d(in_channels=3, out_channels=16, kernel_size=3, padding=0),
        nn.ReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2),
        nn.Conv2d(in_channels=16, out_channels=32, kernel_size=3, padding=0),
        nn.ReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2),
        nn.Flatten(),
        nn.LazyLinear(256),
        nn.ReLU(),
        nn.Linear(256, num_classes)
        )
# YOUR CODE ENDS HERE


@d2l.add_to_class(CNNClassifier_v3)
def configure_optimizers(self):
    return torch.optim.Adam(self.parameters(), self.lr)

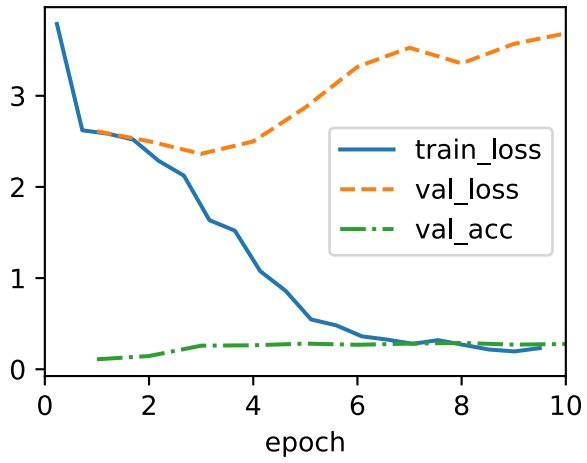
CNNClassifier_v3(num_classes=14).layer_summary((64, 3, 224, 224))
```

```
Conv2d output shape:     torch.Size([64, 16, 222, 222])
ReLU output shape:       torch.Size([64, 16, 222, 222])
MaxPool2d output shape:  torch.Size([64, 16, 111, 111])
Conv2d output shape:     torch.Size([64, 32, 109, 109])
ReLU output shape:       torch.Size([64, 32, 109, 109])
MaxPool2d output shape:  torch.Size([64, 32, 54, 54])
Flatten output shape:    torch.Size([64, 93312])
Linear output shape:     torch.Size([64, 256])
ReLU output shape:       torch.Size([64, 256])
Linear output shape:     torch.Size([64, 14])
```

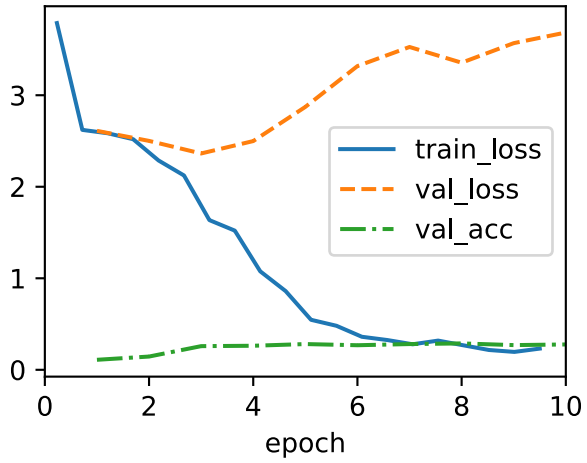
```
# Instantiate the model and trainer, and fit using the PC_Data.

# YOUR CODE STARTS HERE
modelv3 = CNNClassifier_v3(num_classes=14, lr=0.001)
model_trainerv3 = d2l.Trainer(max_epochs=10)
model_trainerv3.fit(modelv3, PCPartsDataImpl)

print_model_results(modelv3.board)
# YOUR CODE ENDS HERE
```




```
========================================
Final Training Loss:     0.2317
Final Validation Loss:   3.6853
----------------------------------------
Final Validation Accuracy: 27.70%
========================================
```



- **Q5.4:** A slightly modernized version of `AlexNet()` is provided below. What is new in this implementation compared to the prior networks you've explored? Be specific.
- **A5.4:** The most clear new thing is the Dropout in the linear layers. It also does multiple convolutions in a row without maxpool in between. Its also a very deep model (hence it being a DNN). Otherwise it still used conv2d, relu, maxpooling, and linear layers.

Also saying it didnt perform tremendously is a gross understatement. We ended up with 28.8% val acc which is better than random guessing but still pretty bad.

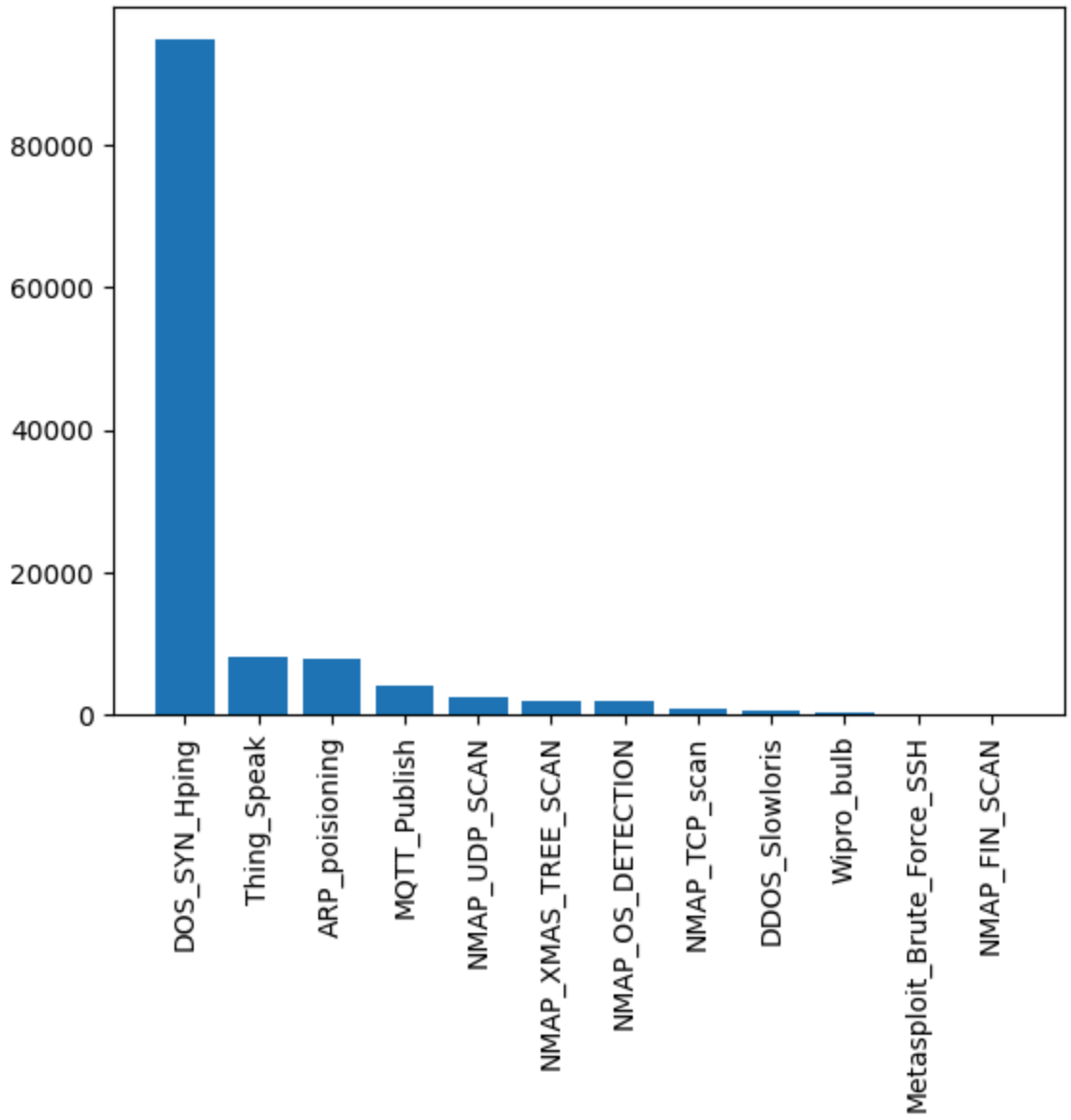
- **Q5.5:** If you ran the above code (or just take our word for it), you will conclude that this AlexNet model, which was considered state-of-the-art not too long ago, still struggles in our environment to accurately classify images of common PC parts. Attempt to explain what might be happening. For this question, you may be able to reason about it directly, but you can also search and learn more about AlexNet. Cite any sources you used.
- **A5.5:** It could be the model, considering even though it was SOTA not that long ago it was still not a great classifier overall. In addition to this the PCParts dataset we are using is very dirty with many very poor images (like Linus Sebastion from Linus Tech Tips for the "headphones" label). I'm not sure how much better it would do if it had a much higher quality dataset, but even then it might also be struggling because the original Imagenet dataset was very large at over 1 million images: https://en.wikipedia.org/wiki/ImageNet.

- **Q5.6:** Why do many of the filters in the first convolutional layer look similar? What does this tell you about the role of early layers in a CNN?
- **A5.6:** The filters from the first layer are likely extracting similar high level features (like edges) which makes it easier for more specialized kernels to properly do more in-depth feature extraction. Its like pre-processing the data for later layers.

- **Q5.7:** EfficientNet and AlexNet have similar levels of complexity, but EfficientNet (using transfer learning) trains much faster. Consider how the model is initialized and what parameters are being updated. Why does this lead to reduced training time?
- **A5.7:** Looking at it, there is an additional dropout layer, and according to geeks for geeks dropout reduces the amount of co-adaption (neurons learning the same feature) which by reducing thiss co-adaption you are doing less redundant calculations. This is likely pretty small to the impact of having pre-trained feature extraction layers, meaning when we do training it is only on the last dense layers for classification rather than all the Conv layers.

```
# YOUR CODE STARTS HERE
counts = df['Attack_type'].value_counts()
plt.bar(counts.index, counts.values)
plt.xticks(rotation=90)
```

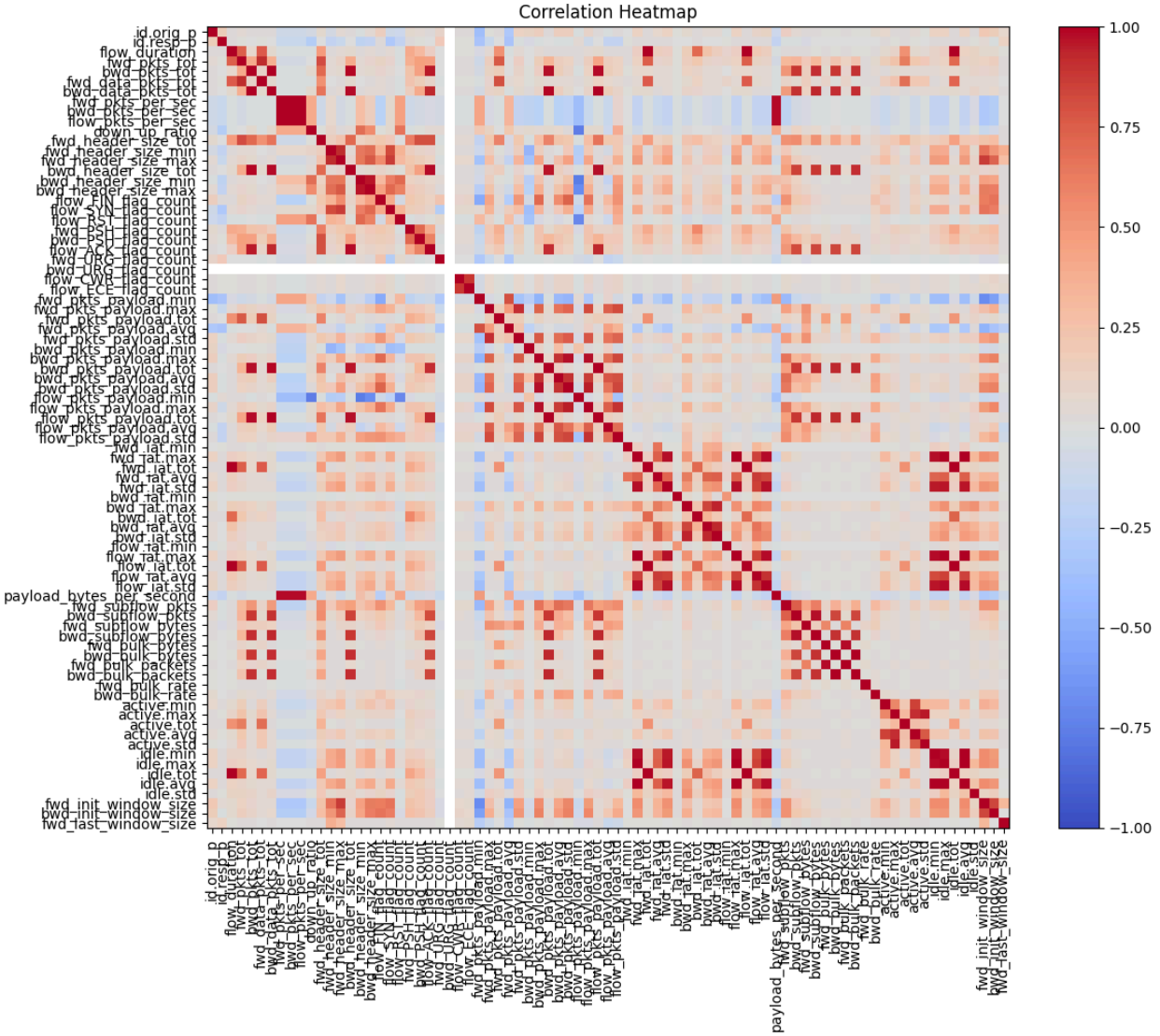
```
plt.show()
# YOUR CODE ENDS HERE
```



- **Q6.1:** Looking at the class distribution, it is clearly heavily imbalanced. If you measure your model's accuracy purely on accuracy, you could hypothetically just guess `DOS_SYN_Hping` every time and get over 70% accuracy. With an imbalanced dataset like this, how could we analyze our model's performance other than just using the raw training / validation accuracy number?
- **A6.1:** We could normalize the dataset to reduce the bias of this class (i.e. drop DOS_SYN_Hping or duplicate others), although that would come at the cost of significantly reducing our dataset size or overfit other examples... Alternatively without it we can do direct sampling rather than pure random but that might have similar drawbacks..

```
# YOUR CODE STARTS HERE
corr = numeric_df.corr()

plt.figure(figsize=(12, 10))
plt.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar()
plt.xticks(ticks=range(len(corr.columns)), labels=corr.columns, rotation=90)
plt.yticks(ticks=range(len(corr.columns)), labels=corr.columns)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
# YOUR CODE ENDS HERE
```




```
from numpy.core.numeric import indices


class CyberDataset(d2l.DataModule):
    def __init__(self, full_df, batch_size=64, train_val_split=0.8):
```

```
        super().__init__()
        self.save_hyperparameters()

        # Go step-by-step here, and debug as you go.
        full_df = full_df.drop(columns=['no'])  # Drop the 'no' column if it
exists

# YOUR CODE STARTS HERE



        # An example of using torch.nn.functional.interpolate to upscale the
9x9 images to 64x64.
        #X_upscaled = torch.nn.functional.interpolate(X_tensor, size=(64, 64),
mode='bilinear', align_corners=False)

        #randomize dataset
        full_df = full_df.sample(frac=1.0,
random_state=42).reset_index(drop=True)
        # Extract and encode labels
        y_indices, self.classes = pd.factorize(full_df['Attack_type'])
        self.y = torch.tensor(y_indices, dtype=torch.long)
        # normalize input features features
        X_data = full_df.select_dtypes(include=['float64', 'int64'])
        X_norm = (X_data - X_data.mean()) / X_data.std().replace(0, 1.0)
        # Reshape 81 features to 1 channel 9x9 grid
        X_tensor = torch.tensor(X_norm.values,
dtype=torch.float32).reshape(-1, 1, 9, 9)
        # Upscale from 9x9 to 64x64 (as described above)
        self.X = torch.nn.functional.interpolate(X_tensor, size=(64, 64),
mode='bilinear', align_corners=False)

        # Allocate training and validation sizes. Adjust this code if your
outputs are named differently.
        num_samples = len(self.y)
        self.num_train = int(num_samples * train_val_split)
        self.num_val = num_samples - self.num_train

# YOUR CODE ENDS HERE
    def get_dataloader(self, train):
        i = slice(0, self.num_train) if train else slice(self.num_train, None)
        return self.get_tensorloader((self.X, self.y), train, i)
```

```
/tmp/ipykernel_6483/4131899939.py:1: DeprecationWarning: numpy.core.numeric is
deprecated and has been renamed to numpy._core.numeric. The numpy._core
namespace contains private NumPy internals and its use is discouraged, as
NumPy internals can change without warning in any release. In practice, most
real-world usage of numpy.core is to access functionality in the public NumPy
API. If that is the case, use the public NumPy API. If not, you are using
NumPy internals. If you would still like to access an internal attribute, use
numpy._core.numeric.indices.
  from numpy.core.numeric import indices
```

- **Q6.2:** Briefly explain what each line of code that you added to `__init__` above is doing. Be specific.
- **A6.2:** The first line with `full_df` is taking and randomizing the full dataset. The next two lines first extract the labels from each data example. The next two lines take the feature inputs and normalize it. Then we reshape the data into a 9x9 grid. Finally using the interpolate function from torch as provided above we upscale from 9x9 to 64x64

- **Q6.3:** As mentioned above, upscaling from 9×9 to 64×64 does not add new information to the data. What are the potential drawbacks of this approach?
- **A6.3:** Because this doesn't meaningfully add additional data it at best does not hurt the examples for learnability. However, by increasing the size we are increasing the computational burden for inference as a 9x9 is cheaper to compute than a 64x64, although in this case it is beneficial as we use pre-trained architectures.

- **Q6.4:** Does the training behavior appear reasonable? Explain why or why not. Note that the model achieves relatively high accuracy after the first epoch. What might explain this? Finally, beyond accuracy, what other methods or metrics could you use to confirm that the model is genuinely learning meaningful patterns?
- **A6.4:** The training behavior looks reasonable curve wise but the accuracy starting very high is concerning. It has high accuracy because, well, 70% of the examples are one class so its easy to fit that class. To verify its actually learning a pattern here it would be good to normalize the classes in the validation so that particular class isn't so overrepresented to ensure it is learning good patterns. Even with the unbalanced data, we have a 98.4% accuracy which is still impressive.

Prelab Questions

- **Q0.1):** Explain the difference between a linear model and a model with nonlinear activation functions. Why are nonlinearities necessary for image classification tasks?

- **A0.1):** A linear model creates a single linear hyperplane to seperate data, a model with nonlinear activations is able to find patterns in nonlinear data (i.e. more complex than a single line). This is generally required for image classification as images require pretty intensive feature extraction, and while manual methods exists these features can be learned by a model. Images create a dataset that isn't really linearlly seperable.

- **Q0.2):** Why is pooling used in convolutional neural networks? What problem does it address, and how does it affect the spatial dimensions of the data?
- **A0.2):** Pooling is used in CNNs because it reduces the activations between layers, which cuts down the input activations (i.e. reduces spatial dims) to the next layer which is important for the final fully connected layer as well as reducing computation.

- **Q0.3):** An input image has shape $32 \times 32 \times 3$. After applying a convolutional layer with a 3×3 kernel, no padding, a stride of 1, and 16 output channels, what is the shape of the output? Show your reasoning.
- **A0.3):** 30 x 30 x 16. This is because there are 16 output channels (last dim), the 3x3 kernel will effectively reduce the size of edges by 1 on all sides (3x3 has 1 pixel on the edges).